$\frac{87}{100}$

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**

Question Count: 5

Code of Conduct Shanika (192511387)

I Beffy . A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Beffy

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

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1. Evolution from standalone computing to cloud - Based systems.

Standalone computing:

* Each computer worked independently
* Data and software were stored locally.
* No resoure sharing between computers
* High maintenance cost
* Limited storage and processing power.

Client - Server computing:

* clients requested services from a central server
* Data stored in centralized servers
* Improved resource sharing
* Easier management and security.

Distributed computing:

* Multiple computers worked together over a network.

* Tasks were divided among systems.

* Increased performance and reliability

Grid computing:

* Combined resources from different locations

* Solved large scientific and engineering problems

* Improved utilization of computing resources.

Cloud computing:

* Provides computing resources over the internet

* users access servies on demand

* pay - as - you - use model

* Highly scalable and flexible

Key Technological Transitions:

* Development of high - speed internet

* Introduction of virtualization

* Large - scale data centers.

Working of Hypervisors in virtualization

Hypervisor :

* Software that creates and manages virual machines
* Allows multiple operating systems to runs on one physical computer
* Allocates cpu , memory , storage and network resources
* provides isolation between virtual machines
* Improved hardware utilization and reduces cost

Type 1 Hypervisor (Bare-Metal)

* Runs directly on physical hardware
* No host operating system required
* High performance
* more secure and reliable
* used in cloud data centers.

Eg:

VMware ESXi , Microsoft Hyper-V , Xen server

Type 2 Hypervisor (Hosted)

* Runs on a host operating system
* Easier to install and use
* Lower performance than Type 1.
* Suitable for personal use and testing.

Eg:

Oracle VirtualBox, VMware Workstation, Parallels Desktop

Difference:

Type 1	Type 2
Runs on hardware	Runs on host OS
Faster	Slower
More secure	Less secure
Enterprise use	Personal / testing use

3. Role of APIs in cloud service integration

API (Application Programming Interface):

* Set of rules for communications between applications.

* Enables interaction between cloud services.

Role of APIs:

* connect cloud applications
* Exchange data between services
* Automate cloud operations
* Access cloud resources programmatically
* Support application development.

APIs Support Interoperability By:

* connecting services from different cloud providers

* Allowing different applications to communicate.

* Enabling secure data sharing
* Supporting REST and SOAP standards
* Integrating public, private, and hybrid clouds.

Benefits :

* Faster Integration
* Reduced development time
* improved flexibility
* Better scalability
* Easy automation.

Security features of APIs:

* use authentication and authorization mechanisms
* Protect data using encryption
* control access through API keys and tokens.

4. Resource Pooling in cloud computing:

Resource pooling:

* Resource pooling is one of the essential charactersis of cloud computing

* It refers to combining computing resources into a shared pool.

* Resource include servers, storage, memory, processors, and networking

* Resource are dynamically assigned and reassigned based on user demand.

* Virtualization technology is used to manage and allocate resources efficiently.

* Users do not know the exact physical location of the resources.

Working of Resource pooling:

* A cloud provider maintains a large pool of physical resources.

* Adds resources during high demand.

* Removes resources during low demand.

* Ensures uninterrupted service

Benefits :

* Handle sudden traffic increases
* improved application performance
* Reduces operational cost
* Efficient resource utilization
* High scalability
* Better customer satisfaction

Advantages :

* cost - effective solution
* Better resource management
* Faster service delivery
* supports business growth

5. Rapid Elasticity in cloud computing

Rapid Elasticity :

* Ability to increase or decrease automatically.
* Resources are allocated according to workloads
* supports dynamic and changing demand
* uses virtualization and automation

Working :

* monitors resource usage continuously
* Adds resources during high demand
* Removes resources during low demand
* Ensures uninterrupted service.

Benefits :

* Handles sudden traffic increases
* Improves application performance

* Reduce operational cost
* Efficient resource utilization
* high scalability
* Better customer satisfaction
* pay only for resources used.

Characteristics of Rapid Elasticity

* Resources can be scaled up or down automatically
* Responds quickly to changing workload demand
* supports both vertical and horizontal scaling
* minimizes resoures wastage
* Ensures continuous availability of cloud services
* Enables the pay - as - you - use pricing model.